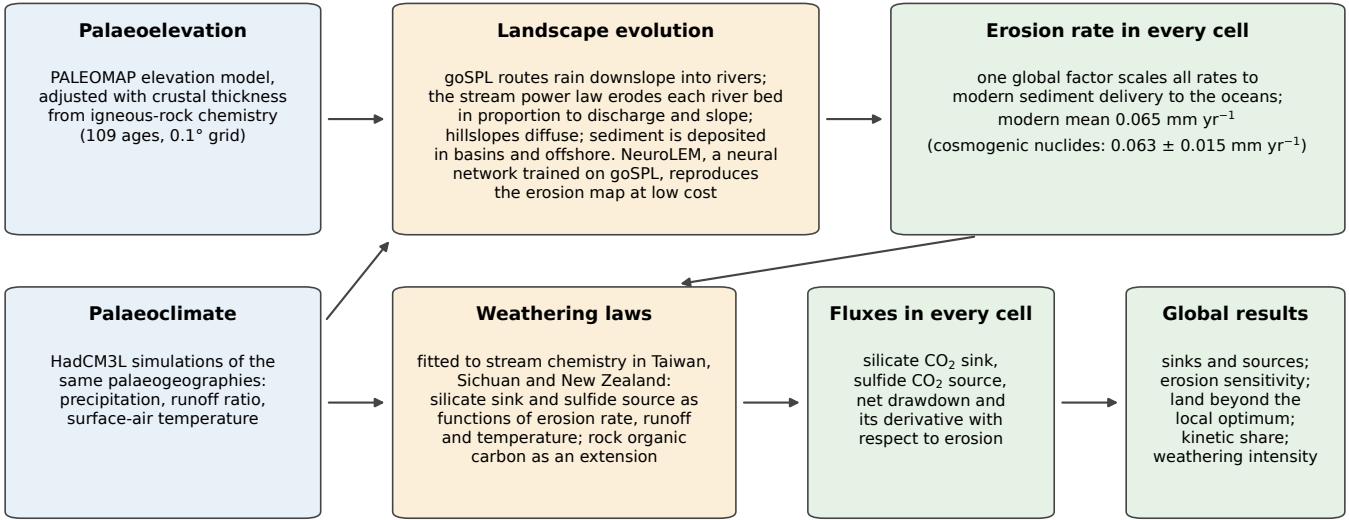
Palaeoelevation
PALEOMAP elevation model, adjusted with crustal thickness from igneous-rock chemistry (109 ages, 0.1° grid)
Landscape evolution
goSPL routes rain downslope into rivers; the stream power law erodes each river bed in proportion to discharge and slope; hillslopes diffuse; sediment is deposited in basins and offshore. NeuroLEM, a neural network trained on goSPL, reproduces the erosion map at low cost
Erosion rate in every cell
one global factor scales all rates to modern sediment delivery to the oceans; modern mean 0.065 mm yr^{-1} (cosmogenic nuclides: 0.063 ± 0.015 mm yr^{-1})
Palaeoclimate
HadCM3L simulations of the same palaeogeographies: precipitation, runoff ratio, surface-air temperature
Weathering laws
fitted to stream chemistry in Taiwan, Sichuan and New Zealand: silicate sink and sulfide source as functions of erosion rate, runoff and temperature; rock organic carbon as an extension
Fluxes in every cell
silicate CO_2 sink, sulfide CO_2 source, net drawdown and its derivative with respect to erosion
Global results
sinks and sources; erosion sensitivity; land beyond the local optimum; kinetic share; weathering intensity